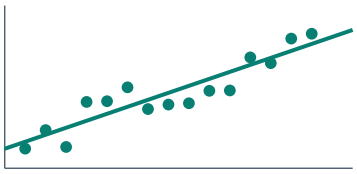
IBEX / MACHINE LEARNING STUDIO

Machine learning, visually

Choose the task → prepare the data → train → evaluate → predict and save.

1 Regression

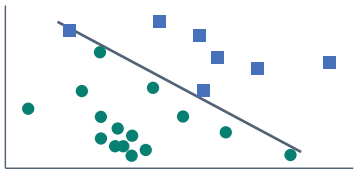
Predict a number



Start: **Linear Regression**
Curvature? Try polynomial or trees.

2 Classification

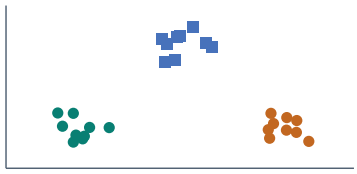
Predict a category



Start: **Logistic Regression**
Shapes show two class labels.

3 Clustering

Discover groups; no target



Start: **K-Means** for compact groups.
Density + noise? Try DBSCAN.

Import CSV → **Inspect** → **Prepare** → **Train** → **Evaluate** → **Save**

Scale before distance-based learning

Different units can dominate similarity.

Raw feature ranges

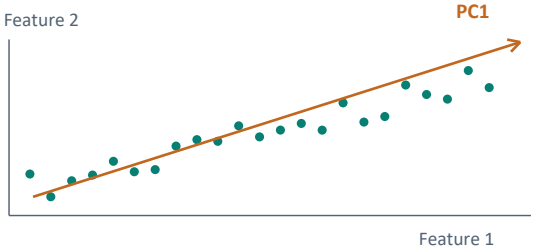


ZScore = (x - mean) / SD
MinMax = (x - min) / (max - min)
MaxAbs = x / max(|x|)

Useful for KNN, SVM, neural nets and distance-based clustering.
Keep category-code mappings.

PCA: a new view of the features

Reduce dimensions; retain the original input meaning.



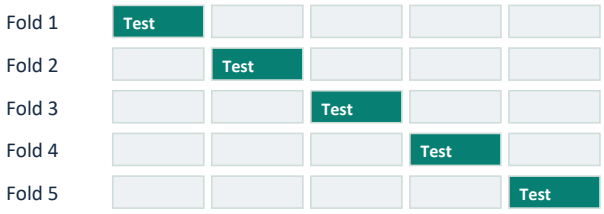
PC1 follows the greatest feature variation. Subsequent components capture remaining variation.

Preserve the target with **Last column is target (Y)**. Predict using original features when the model stores its PCA recipe.

95% variance retained ≠ 95% prediction accuracy.

K-fold: every row tested once

Example: 5 folds; shaded block = held-out test rows.

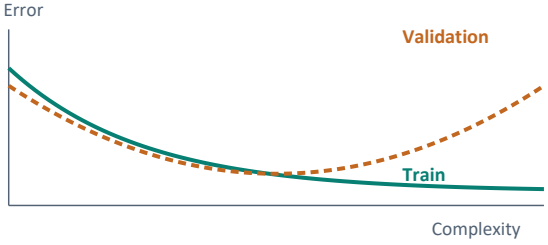


Combined accuracy: total correct held-out predictions / total held-out rows.
Combined R^2: calculated from pooled predictions; not the mean fold R^2.

Training counts repeat rows across folds. Read the Combined **test** row for held-out results.

Recognize overfitting

Compare training and validation performance.



Train error falls while validation error rises? Reduce complexity, inspect leakage and seek more useful data.

IBEX: preprocessing is not refitted inside folds. Splits are random, not time-, group- or class-aware. Scores are exploratory.

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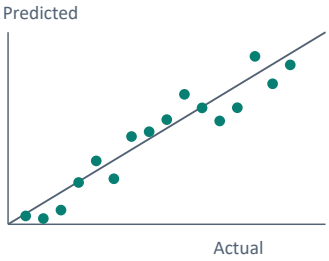
A4 | Print at 100% | Illustrative diagrams, not measured results | September 2026

Read the plots. Choose the model.

Use several diagnostics together; one impressive score is not enough.

Regression: actual vs predicted

Points near the diagonal indicate closer predictions.

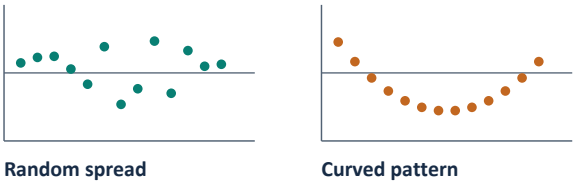


R^2 ↑
1: perfect
0: mean baseline
< 0: worse

MAE ↓ = mean |actual - predicted|
RMSE ↓ = √mean(error²); penalizes large errors.
Error units follow the evaluated target scale.

Residuals: inspect the pattern

Residual = actual - predicted in these illustrations.



Left: no obvious structure.
Right: the model may miss a nonlinear trend.
A fan-shaped spread suggests changing error variance.

Classification: confusion matrix

Rows = actual; columns = predicted.

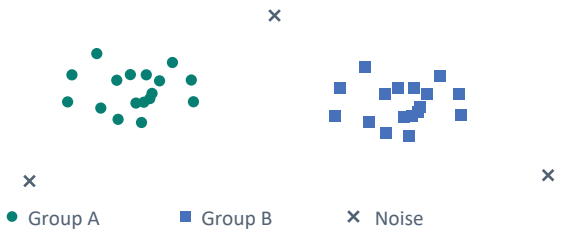
	Predicted: Yes	Predicted: No
Actual Yes	TP 40	FN 10
Actual No	FP 5	TN 45

Example:
Accuracy 85%
Precision 88.9%
Recall 80%

Precision = TP / (TP + FP); recall = TP / (TP + FN).
F1 = 2PR / (P + R). Macro F1 weights classes equally.
Accuracy can hide poor minority-class recall.

Clustering: groups and noise

Choose features that define meaningful similarity.



DBSCAN: can mark noise as -1. K-Means assigns every row to a center. **Silhouette:** higher usually means better separation; not useful with one cluster.

ALGORITHM MAP / start simple, then compare held-out results

11 regressors

Linear Regression · rate, iterations, L1/L2
Polynomial · degree; powers, no interactions
Normal Equation · ridge stabilization
Locally Weighted · tau bandwidth, lambda
Random Forest · trees, depth, minLeaf
Neural Net (MLP) · layers, size, epochs, rate
RBF Surrogate · epsilon width, smoothing
FastTree · trees, leaves, rate
LightGBM · iterations, leaves, rate
Online Gradient Descent · iterations, rate
XGBoost · rounds, depth, eta, subsample

9 classifiers

Logistic Regression · linear baseline
K-Nearest Neighbors · K; scale matters
Naive Bayes · Gaussian baseline
SVM (linear) · C; no RBF kernel
Random Forest · trees, depth, minLeaf
FastTree · boosted one-vs-all trees
LightGBM · trainer defaults in IBEX
XGBoost · rounds, depth, rate
Neural Net (ANN) · layers, size, iterations, rate

Some shared controls are unused by specific classifiers. Check the full guide before tuning.

4 clusterers

K-Means · K, iterations; nearest center
DBSCAN · Eps radius, MinPts density
Hierarchical · K; single / average / complete
Fuzzy C-Means · K, fuzziness M > 1

Before trusting a result
Remove leakage; check class balance and data ranges. Keep a separate final evaluation dataset.

In IBEX
F1 opens help. Save tables, charts and your model. Reset page does not undo preparation.

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